

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

8th Semester of B. Tech. Examination (EE)

May 2014

EE407 Commissioning of Electrical Equipments

Date: 07.05.2014, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Define routine test, type test and special test. [03]
- (b) Write a technical note on Effects of electric current on human body. [04]
- Q - 2 (a) After sudden rain fall in summer the insulation resistance of power transformer is measured and it is 100Ω . As a substation in charge, what will your actions? Do explain in detail. [07]
- (b) Explain the procedure for power frequency voltage withstand test and impulse voltage withstand test of power transformer. [07]

OR

- Q - 2 (a) List the desired characteristics of insulating oil and testing of insulating oil by flash point test, crackle test and sight test. [08]
- (b) Explain the term “Induction motor stator and rotor interaction” [06]
- Q - 3 (a) Discuss causes, consequences and remedies for following abnormal conditions in DC machine: 1) Generator fails to build up its voltage, 2) Sparking at brushes, 3) Heat in generator, 4) Noise [08]
- (b) Explain the difference between earthing and grounding. With necessary diagram explain and give comparison of reactance earthing and resistance earthing. [06]

OR

- Q - 3 (a) Enlighten the process of carrying out bar to bar test for d.c. machine's trouble shooting? What is its purpose? [06]
- (b) How will you check the correct polarity & terminal marking of Current Transformer? [04]
- (c) On site, if no supply is available and you have only galvanometer and you are suppose to measure the speed of three phase induction motor, what will you do and how? (Name plate of machine is also missing & tachometer is not available) [04]

SECTION – II

- Q - 4**
- (i) What do you expect to happen if, full rated current is applied to a power transformer while performing open circuit test? [07]
 - (ii) Can Dy_1 type of transformer be connected in parallel with Dy_{11} type? Justify your answer with relevant connection and vector diagram.
 - (iii) PI is a ratio of dielectric absorption over a ____ min period to ____ min period.
 - (iv) List out mechanical tests to be performed on electrical equipments before commissioning.
 - (v) Name the various methods to find out correct phase sequence before connecting in parallel operation.
 - (vi) What will happen if reversal of one phase winding in case on induction motor?
 - (vii) Cogging
- Q - 5 (a)** Explain the following troubles in Synchronous machine. 1) Motor fails to start; [08]
2) Motor does not attain proper speed; 3) Motor fails to pull into step; 4) Motor trips the breaker or pulls out of step.
- (b)**
- (i) What do you understand by a negative regulation of an alternator? [06]
 - (ii) A slight decrease in field current worsens the operating power factor. Was the power factor leading or lagging?
 - (iii) "Motor hunting" – give causes and remedies.

OR

- Q - 5 (a)** Write down the steps for installation and commissioning of DC machine. [08]
- (b)** Explain the following term and their remedies: [06]
- (i) Bearings overloaded
 - (ii) Motors starts and stops.
 - (iii) Motor giving noise
- Q - 6 (a)** State the type of fault and the most efficient method of test for locating that type of fault in an underground cable. Explain any one method in detail. [07]
- (b)** State the possible defects and remedies in the various equipments of a power line. [07]

OR

- Q - 6 (a)** State and explain the various tests necessary on CT. [07]
- (b)** Prepare the maintenance chart of Air circuit breaker. [07]
